

Incidence of Childhood Stroke and Association With Recent Infection

A Population-Based Study Using Linked Data

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Neurology® 2026;106:e214704. doi:10.1212/WNL.00000000000214704

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Abstract

Background and Objectives

There are limited population-based data on the incidence and risk factors for childhood stroke, including associations with recent infection and vaccination. Therefore, we aimed to determine (1) the incidence of childhood ischemic and hemorrhagic stroke and (2) associations between a recent infection or vaccination and the risk of childhood stroke.

Methods

A population-based case-control study was undertaken using linked administrative data from 2017 to 2023 in Victoria, Australia. All residents, aged 28 days to <18 years, were included. Infections in the 60 days immediately before admission were ascertained using state-wide notifiable infections and hospital admissions data sets. Vaccinations in the 42 days immediately before admission were ascertained using the Australian Immunisation Register. Strokes were identified in state-wide hospital admissions and death registry databases. Each case of childhood stroke was matched with up to 5 nonstroke controls on age (1-year bands), sex, socioeconomic position, comorbidities, and year. Conditional logistic regression was used to examine associations between infection and vaccination with childhood stroke.

Results

Over the 7 years, 571 childhood strokes (60% ischemic; median age 8.2 years; 44% female patients) occurred among 9,881,934 person-years (5.8 events [95% CI 5.3–6.3] per 100,000 person-years). Two in 5 patients had evidence of an infection in the preceding 60 days, with lower respiratory infections ($n = 74$), sepsis ($n = 67$), upper respiratory tract infections ($n = 63$), meningitis/encephalitis ($n = 40$), gastrointestinal tract infections ($n = 28$), and urinary tract infections ($n = 22$) being the most common. Children with stroke (vs matched controls) were more likely to have had a recent infection (cases 41.9% vs controls 18.6%; adjusted odds ratio [aOR] 2.54, 95% CI 1.97–3.27). The magnitude of this association was greater for ischemic stroke (aOR 3.39, 95% CI 2.45–4.69) than hemorrhagic stroke (aOR 1.90, 95% CI 1.24–2.90). No significant association was detected between recent vaccination and childhood stroke (cases 4.2% vs controls 2.5%; aOR 1.69, 95% CI 0.92–3.09).

Discussion

We provide contemporary data on new cases of childhood stroke in Australia, using population-wide linked data. Having a recent infection doubled the risk of childhood stroke. We found no significant evidence of an association between vaccination and childhood stroke.

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Glossary

COVID-19 = coronavirus disease 2019; **ICD-10-AM** = International Statistical Classification of Diseases and Related Health Problems, Tenth Revision, Australian Modification; **OR** = odds ratio; **RSV** = respiratory syncytial virus; **VIPS** = Vascular Effects of Infection in Childhood Stroke; **VSHL** = Vaccine Safety Health Link.

Introduction

Childhood stroke is associated with significant mortality, morbidity, and long-term financial cost burden.^{1,2} Data from systematic reviews suggest that childhood stroke is rare, with an estimated 0.9–7.9 ischemic and 0.1–3.2 hemorrhagic events occurring per 100,000 persons-years.^{3,4} However, contemporary data on childhood stroke event rates are lacking, particularly within Australia.

Unlike the adult population, children with stroke do not typically present with traditional risk factors, such as hypertension and dyslipidemia.⁵ Rather, childhood stroke tends to be driven by other factors, such as congenital heart defects, nonatherosclerotic arteriopathies, genetic diseases, and blood clotting disorders.¹ There is also growing evidence that childhood strokes can be triggered by a range of infections including sepsis, meningitis, endocarditis, influenza, coronavirus disease 2019 (COVID-19), and varicella zoster.^{6,7}

In the Vascular Effects of Infection in Childhood Stroke (VIPS) study,⁸ a case-control study across 6 countries (N = 355 cases and 354 stroke-free controls), 18% of children had an infection in the week before stroke, with upper respiratory infections the most common.⁸ VIPS investigators identified an increased risk of stroke in children with fewer or no vaccinations vs those who received all or most vaccinations (odds ratio [OR] 7.3, 95% CI 2.5–21).⁸ In a separate case-control study, childhood stroke was reported after varicella zoster vaccination,⁹ although this association was subsequently invalidated in a Canadian population-based study.¹⁰ Additional data on the role of specific infections and vaccinations with childhood stroke are needed.

Because most literature on this topic to date has relied on hospital-based recruitment and self-reported exposure data, there is a need for comprehensive, population-wide data to better understand the epidemiology of childhood stroke, including the role of infections and vaccinations. Therefore, we aimed to determine the population-level incidence of childhood stroke and whether recent infections and vaccinations were associated with childhood stroke.

Methods

Study Design and Setting

A population-based observational study was undertaken using linked administrative health data from Victoria, Australia

(2017–2023)—the second most populous state, with 6.9 million residents (1.4 million children; at June 30, 2023).¹¹ The reporting of this study follows the REporting of studies Conducted using Observational Routinely-collected health Data statement.¹²

Data Sources

Data were accessed from the Vaccine Safety Health Link (VSHL),¹³ an enduring linked data asset established in 2021 to strengthen existing vaccine surveillance methods.¹³ VSHL includes data from all permanent residents of Victoria (irrespective of vaccination status) and temporary residents who have a vaccination record in Australia. This study involved the analysis of VSHL data, linked at the person-level with 98% accuracy,¹³ from the following sources:

- Australian Immunisation Register: established in 2016, includes data (e.g., vaccine code and date) on all vaccinations administered within Australia, including those government funded or purchased privately. From 1996 to 2016, this database was specifically a childhood register (Australian Childhood Immunisation Register), capturing information on all vaccines administered to children aged <7 years.¹⁴ Since late 2018, state/territory health departments have provided historical and prospective data on all vaccines given to children aged 7–18 years as part of school-based immunization programs.¹⁴ VSHL contains the entire Australian Immunisation Register for Victorian residents with vaccine data available for all participants of this study since birth.
- Victorian Admitted Episode Data Set includes data relating to inpatient admissions at all public and private hospitals in Victoria (e.g., admission and discharge dates, International Statistical Classification of Diseases and Related Health Problems, Tenth Revision, Australian Modification [ICD-10-AM] diagnoses).
- Victorian Emergency Minimum Data Set includes data (e.g., arrival and separation dates, ICD-10-AM diagnoses) relating to presentations to public emergency departments in Victoria.
- Victorian Death Index includes data (date of death, underlying, and associated causes of death) on all deaths (over 20 weeks of gestation or ≥400 g) occurring in Victoria.
- Public Health Event Surveillance System includes data on 76 notifiable conditions reported by health care providers or laboratories under legislation (including vaccine preventable infections such as influenza,

respiratory syncytial virus [RSV], pneumococcal, and varicella zoster).¹⁵

- Transmission and Response Epidemiology Victoria data set includes information on all COVID-19 tests conducted in Victoria. It also includes COVID-19 case notifications made by health care providers, laboratories, and patients.

Standard Protocol Approvals, Registrations, and Patient Consents

Approval for the VSHL project was obtained from The Royal Children's Hospital Melbourne Ethics Committee (HREC approval 79964). Informed consent was waived by this committee for the analysis of deidentified patient data. Specific HREC approval for this study was also obtained from Monash University (approval 39123). The study was conducted in accordance with the Declaration of Helsinki.

Cases of Stroke

The outcome was the first occurrence of a childhood stroke from 28 days to <18 years of age. Using a previously validated ICD-10-AM code list in Australia,^{16,17} we identified episodes of ischemic (I63, I64, I65.2, I66.0-I66.2, I66.8, I66.9, I67.5, I67.8, I67.9, I69.3, and I69.4) or hemorrhagic (I60, I61, and I62) stroke recorded as a principal or additional diagnosis (codes outlined in eTable 1). Where multiple records of stroke existed for the same person during our observation period, only the first record was included as the index stroke event. Cause of death fields were also searched to identify any nonadmitted, fatal episodes of childhood stroke (keywords outlined in eTable 1).

Variables

Demographics (e.g., age, sex, or country of birth) were derived using hospital admissions data. Where this information was unavailable (e.g., for fatal strokes), data were imputed from the immunization register, or from previous hospitalizations, for the same patient. Socioeconomic position was based on the Index of Relative Socioeconomic Advantage and Disadvantage 2021,¹⁸ a validated neighborhood-level measure derived using residential postcodes. A previously validated algorithm was applied to derive common pediatric comorbidities using ICD-10-AM diagnoses for all hospital admissions before, and including, the index event (code list in eTable 2).¹⁹ These comorbidities were also summarized into a single weighted pediatric comorbidity index that is predictive of 1-year hospitalizations.¹⁹

Recent infections before the index event were identified using the notifiable infection registers and the hospital administrative data sets (admissions and emergency; codes provided in eTable 3). A 60-day look-back period was chosen to align with previous epidemiologic definitions used by the US Centers for Disease Control and Prevention to detect COVID-19 using hospital administrative data.²⁰ This look-back period is also supported by a previous case-control study conducted in adults (N = 36,975) where associations between influenza-

like infection and stroke were attenuated beyond 60 days after infection.²¹ Recent vaccinations in the 42 days before the index event were identified using vaccination records in the immunization register. This look-back period was chosen to align with previous vaccine safety studies²² and increases the confidence by which a recent vaccination can be plausibly related with the outcome. Additional classification was undertaken to categorize common childhood vaccines (codes in eTable 4).

Incidence of Childhood Stroke

Incidence rates of stroke overall, and by type (ischemic and hemorrhagic), were estimated for the entire childhood population of Victoria. Numerators comprised cases of childhood stroke (aged 28 days to 18 years) identified in the linked administrative data. Denominators were drawn from the Australian Bureau of Statistics Estimated Resident Population of Victoria at the midpoint of each year (i.e., June 30).¹¹ Age standardization was undertaken to more reliably compare rates over time. This involved the direct standardization method, using 1-year age bands, to the 2001 Australian Bureau of Statistics Standard Population (most up-to-date version of the standard population).²³ All rates were expressed per 100,000 person-years.

Matched Case Control

To examine factors associated with childhood stroke, each case of childhood stroke was matched with up to 5 stroke-free controls admitted to hospital. Matching was undertaken based on age (1-year bands), sex (male vs female), quintile of socioeconomic position, childhood comorbidity score (0, 1–2, or 3+), and year of admission.

Statistical Analysis

Descriptive statistics were used to summarize the baseline characteristics of patients with and without stroke. Differences between groups were evaluated using χ^2 tests for categorical variables, and Wilcoxon rank-sum tests for nonparametric continuous variables. For the estimation of incidence rates, we obtained exact 95% CIs from the Poisson distribution. For the matched case-control, associations between pre-event factors (infections and vaccinations) and childhood stroke were determined using conditional logistic regression. Additional multivariable adjustment was undertaken to minimize potential residual confounding.²⁴ All data were prepared using Microsoft SQL Server Management Studio and analyzed using Stata/SE 18.0 for Windows (StataCorp., College Station, TX, 2023). A 2-tailed *p* value <0.05 was considered statistically significant. Data were complete for all variables, apart from country of birth (0.01%) and neighborhood socioeconomic quintile (0.1%).

Sensitivity Analyses

Because causal associations cannot be drawn from case-control studies, sensitivity analyses were undertaken to validate any vaccine safety signals using a self-controlled case series design. This robust epidemiologic study design eliminates all sources of

time-invariant confounding because individuals act as their own control (see Supplemental Methods).^{25,26} We also confirmed that the results for recent vaccination were robust using shorter risk windows of 14 and 28 days before stroke. A sensitivity analysis was also undertaken to examine whether a time-dependent dose-response relationship existed between recent infection and childhood stroke. For this analysis, we identified infections occurring most proximally to the admission in the following mutually exclusive look-back periods: 1–14, 15–60, 61–120, 121–150, 180–360, and 361–720 days.

Data Availability and Access

All VSHL data are deidentified and only accessible by approved project members in the Centre for Victorian Data Linkage secure platform. To prevent possible reidentification of individual patients, all dates have been systematically perturbed by a maximum period of 2 weeks. Any cell counts comprising <5 individuals have been suppressed to adhere to confidentiality requirements. All analyses were undertaken on deidentified data by the lead author, following approval from relevant data custodians. Because of ethical and legal restrictions, person-level data from this study cannot be shared. However, certain aggregated data and coding that support the findings of this study are available on reasonable request from the corresponding author, following approval from the relevant data custodians.

Results

Over the 7-year period, a total of 571 children with stroke were identified (60% ischemic; 40% hemorrhagic) in Victoria. The median age of children with stroke was 8.2 years (25th–75th percentile 2.0–14.2 years) and approximately half (44%) were female. Compared with patients admitted with nonstroke conditions, children with stroke tended to be 1.1 years younger, from areas of socioeconomic disadvantage, and were more likely to have comorbid conditions (Table 1).

Childhood stroke occurred at a rate of 5.8 events per 100,000 persons per year, with notable variation by age and sex (Table 2). The age-specific incidence of childhood stroke was the greatest among infants aged 28 days to 1 year and was significantly elevated among male individuals than female individuals in this age category. The incidence of childhood stroke seemed to decline steadily between 2 and 13 years, after which a slight increase was observed among 14–17 year old children. Although the incidence of ischemic stroke was greater than hemorrhagic stroke (3.7 vs 2.1 events per 100,000 person years), similar patterns were observed by age and sex. Between 2017 and 2023, the age-standardized incidence of childhood stroke increased by 42% (Table 3).

A total of 571 cases of childhood stroke were matched with 2,734 controls admitted with a nonstroke condition (eFigure 1). After matching, sociodemographic factors and comorbidities were comparable between children with and

without stroke (eTable 5). Of the 571 cases of childhood stroke, 239 (41.9%) had evidence of an infection in the preceding 60 days (39.9% had ≥ 2 infections in the previous 2-year period; Figure 1). Among these infections, the most common were lower respiratory infections ($n = 74$), sepsis ($n = 67$), upper respiratory infections ($n = 63$), meningitis/encephalitis ($n = 40$), gastrointestinal infections ($n = 28$), and urinary tract infections ($n = 22$). The most common upper respiratory infections included severe acute respiratory syndrome coronavirus 2 (COVID-19; $n = 18$), RSV ($n = 13$), influenza virus ($n = 7$), and pneumococcal disease ($n = 8$). Varicella zoster virus occurred in <5 children with stroke and was unable to be investigated further because of privacy concerns arising from this small sample.

Having an infection in the previous 60 days was associated with 2.5-fold increased odds of childhood stroke (OR 2.54, 95% CI 1.97–3.27; Table 4). Further analyses were undertaken to ascertain the association of each major infection with childhood stroke (eTable 6). Specific infections associated with an increased risk of childhood stroke included meningitis/encephalitis (OR 27.36, 95% CI 9.29–80.55), sepsis (OR 9.94, 95% CI 5.02–19.68), lower respiratory infections (OR 3.57, 95% CI 2.26–5.65), gastrointestinal infections (OR 3.44, 95% CI 1.67–7.08), RSV (OR 3.32, 95% CI 1.01–10.85), and other unspecified infections (OR 3.17, 95% CI 2.24–4.49). The association of infection with stroke was time-dependent (Figure 2), with the greatest risk evident for infections occurring most proximal to the event. An elevated risk of ischemic stroke was observed up to 180 days postinfection and up to 60 days postinfection for hemorrhagic stroke.

Of the 571 cases of childhood stroke, 24 (4.2%) had evidence of a vaccination in the preceding 42 days. The most common vaccines received, either in a single or combined dose, were rotavirus ($n = 8$), varicella ($n = 8$), influenza ($n = 7$), and pneumococcal ($n = 7$). In the matched analysis with further direct adjustment for residual confounding (Table 4), there was no evidence of statistically significant increased odds of childhood stroke in patients with a recent vaccination. Similar findings were obtained when using a shorter look-back period of 14 or 28 days (eTable 7) and when examining the association of specific vaccinations with childhood stroke (eTable 8). Although a borderline association between recent influenza vaccination and childhood stroke was initially identified in the matched analysis, this was attenuated after further adjustment and could not be verified in a follow-up self-controlled case series analysis (OR 1.74, 95% CI 0.77–3.96; eTable 9).

Discussion

In this population-based study of >1.4 million children in Australia from 2017 to 2023, we provide contemporary evidence on the incidence of childhood stroke, showing that this

Table 1 Characteristics of Children With and Without Childhood Stroke

	Childhood stroke (n = 571), n (%)	Nonstroke admission (n = 426,432), n (%)	p Value
Female	249 (43.6)	196,046 (46.0)	0.26
Age, categories			
0–1 y	143 (25.0)	47,179 (11.1)	<0.001
2–5 y	99 (17.3)	92,005 (21.6)	
6–9 y	90 (15.8)	87,926 (20.6)	
10–13 y	90 (15.8)	80,810 (19.0)	
14–18 y	149 (26.1)	118,512 (27.8)	
Median age, y (25th–75th percentile) [min–max]	8.2 (2.0–14.2) [0.1–17.9]	9.3 (4.7–14.5) [0.1–17.9]	<0.001
Born in Australia	465 (91.0)	393,447 (92.3)	0.28
Socioeconomic position ^a			
Least advantaged	106 (18.6)	61,282 (14.4)	<0.001
Second least advantaged	103 (18.1)	65,650 (15.4)	
Third most disadvantaged	126 (22.1)	78,197 (18.4)	
Second most advantaged	123 (21.6)	99,738 (23.4)	
Most advantaged	112 (19.7)	121,101 (28.4)	
Type of stroke			
Intracerebral hemorrhage	203 (39.7)	—	
Ischemic stroke	308 (60.3)	—	
Comorbidities ^b			
Anemia	34 (6.0)	2,894 (0.7)	<0.001
Anxiety	32 (5.6)	4,788 (1.1)	<0.001
Cancer	52 (9.1)	1,146 (0.3)	<0.001
Asthma	18 (3.2)	13,307 (3.1)	0.97
Cardiovascular disease ^c	546 (95.6)	10,375 (2.4)	<0.001
Chromosomal abnormality	16 (2.8)	148 (<0.1)	<0.001
Congenital disorder	154 (27.0)	14,007 (3.3)	<0.001
Depression	7 (1.2)	3,149 (0.7)	0.17
Development delay	11 (1.9)	755 (0.2)	<0.001
Diabetes	10 (1.8)	3,428 (0.8)	0.011
Epilepsy	167 (29.3)	6,891 (1.6)	<0.001
Gastrointestinal disease	80 (14.0)	10,779 (2.5)	<0.001
Joint disorder	20 (3.5)	1,725 (0.4)	<0.001
Menstrual disorder	47 (8.2)	1,925 (0.5)	<0.001
Nausea	111 (19.4)	12,016 (2.8)	<0.001
Pain	90 (15.8)	33,763 (7.9)	<0.001
Psychotic	8 (1.4)	846 (0.2)	<0.001
Sleep disorder	50 (8.8)	25,264 (5.9)	0.004

Continued

Table 1 Characteristics of Children With and Without Childhood Stroke (continued)

	Childhood stroke (n = 571), n (%)	Nonstroke admission (n = 426,432), n (%)	p Value
Weight loss	71 (12.4)	1,555 (0.4)	<0.001

^a Measured by Index of Relative Socioeconomic Advantage and Disadvantage.¹⁸
^b Other elements of the childhood comorbidity score (i.e., alcohol, smoking, and drug use) were omitted as the frequency of these conditions were below the minimum output size (≥5) for reporting.
^c Including stroke (I60-I69). Cases identified from the death registry may not have had a previous hospitalization where stroke was coded.

rare event occurs in approximately 1 in every 18,000 children (aged 28 days to <18 years). It is important to note that, a recent infection increased the odds of childhood stroke by approximately 2-fold, and this elevated risk persisted for up to 6 months postinfection. Using complete data on all vaccinations administered since birth, we were unable to detect

Table 2 Incidence of Childhood Stroke in Victoria, Australia, by Stroke Type, Age, Sex, and Overall

	Population at risk (2017–2023), in person-years ^a		Annual incidence rates per 100,000 person-years						Male:female rate ratio Incidence rate ratio (95% CI)	
			Males		Females		All			
	Males	Females	n ^b	Rate (95% CI)	n ^b	Rate (95% CI)	n ^b	Rate (95% CI)		
Combined										
Age										
28 d–1 y	557,368	530,414	89	16.0 (12.8–19.6)	54	10.2 (7.6–13.3)	143	13.1 (11.1–15.5)	1.57 (1.11–2.24)	
2–5 y	1,151,221	1,091,425	53	4.6 (3.4–6.0)	46	4.2 (3.1–5.6)	99	4.4 (3.6–5.4)	1.09 (0.72–1.66)	
6–9 y	1,166,490	1,103,982	48	4.1 (3.0–5.5)	42	3.8 (2.7–5.1)	90	4.0 (3.2–4.9)	1.08 (0.70–1.68)	
10–13 y	1,129,589	1,069,536	45	4.0 (2.9–5.3)	45	4.2 (3.1–5.6)	90	4.1 (3.3–5.0)	0.95 (0.61–1.46)	
14–17 y	1,069,516	1,012,393	87	8.1 (6.5–10.0)	62	6.1 (4.7–7.9)	149	7.2 (6.1–8.4)	1.33 (0.95–1.87)	
All ages	5,074,184	4,807,750	322	6.3 (5.7–7.1)	249	5.2 (4.6–5.9)	571	5.8 (5.3–6.3)	1.23 (1.04–1.45)	
Ischemic										
Age										
28 d–1 y	557,368	530,414	52	9.3 (7.0–12.2)	36	6.8 (4.8–9.4)	88	8.1 (6.5–10.0)	1.37 (0.88–2.16)	
2–5 y	1,151,221	1,091,425	30	2.6 (1.8–3.7)	36	3.3 (2.3–4.6)	66	2.9 (2.3–3.7)	0.79 (0.47–1.32)	
6–9 y	1,166,490	1,103,982	39	3.3 (2.4–4.6)	26	2.4 (1.5–3.5)	65	2.9 (2.2–3.6)	1.42 (0.84–2.43)	
10–13 y	1,129,589	1,069,536	31	2.7 (1.9–3.9)	30	2.8 (1.9–4.0)	61	2.8 (2.1–3.6)	0.98 (0.57–1.67)	
14–17 y	1,069,516	1,012,393	53	5.0 (3.7–6.5)	32	3.2 (2.2–4.5)	85	4.1 (3.3–5.0)	1.57 (0.99–2.51)	
All ages	5,074,184	4,807,750	205	4.0 (3.5–4.6)	160	3.3 (2.8–3.9)	365	3.7 (3.3–4.1)	1.21 (0.98–1.50)	
Hemorrhagic										
Age										
28 d–1 y	557,368	530,414	37	6.6 (4.7–9.2)	18	3.4 (2.0–5.4)	55	5.1 (3.8–6.6)	1.96 (1.09–3.65)	
2–5 y	1,151,221	1,091,425	23	2.0 (1.3–3.0)	10	0.9 (0.4–1.7)	33	1.5 (1.0–2.1)	2.18 (1.00–5.13)	
6–9 y	1,166,490	1,103,982	9	0.8 (0.4–1.5)	16	1.4 (0.8–2.4)	25	1.1 (0.7–1.6)	0.53 (0.21–1.28)	
10–13 y	1,129,589	1,069,536	14	1.2 (0.7–2.1)	15	1.4 (0.8–2.3)	29	1.3 (0.9–1.9)	0.88 (0.40–1.96)	
14–17 y	1,069,516	1,012,393	34	3.2 (2.2–4.4)	30	3.0 (2.0–4.2)	64	3.1 (2.4–3.9)	1.07 (0.64–1.81)	
All ages	5,074,184	4,807,750	117	2.3 (1.9–2.8)	89	1.9 (1.5–2.3)	206	2.1 (1.8–2.4)	1.25 (0.94–1.66)	

^a Based on the total resident population of Victoria at June 30 in each year of the study period (2017–2023), sourced from the Australian Bureau of Statistics.¹¹ 0–1 year population used for 28 days–1 year denominator.
^b Total number of stroke events over the 7-year observation period (2017–2023).

Table 3 Incidence of Childhood Stroke in Victoria, Australia, by Year

Year	Population at risk ^a	n ^b	Crude incidence rate (95% CI), per 100,000 person-years	Age-standardized incidence rate (95% CI), per 100,000 person-years ^c
2017	1,376,052	73	5.3 (4.2–6.7)	5.3 (4.1–6.6)
2018	1,393,932	64	4.6 (3.5–5.9)	4.6 (3.5–5.9)
2019	1,410,255	63	4.5 (3.4–5.7)	4.5 (3.4–5.7)
2020	1,418,825	78	5.5 (4.3–6.9)	5.5 (4.4–6.9)
2021	1,413,313	82	5.8 (4.6–7.2)	5.8 (4.6–7.2)
2022	1,424,269	103	7.2 (5.9–8.8)	7.2 (5.9–8.8)
2023	1,445,288	108	7.5 (6.1–9.0)	7.5 (6.1–9.0)

^a Estimated Resident Population of Victoria at June 30 of the relevant year, sourced from the Australian Bureau of Statistics.¹¹ 0–1 year population used in the absence of more specific data on the 28 days–1 year population.

^b Total number of stroke events over the 7-year observation period (2017–2023).

^c Standardized to the 2001 Australian Bureau of Statistics Standard Population.²³

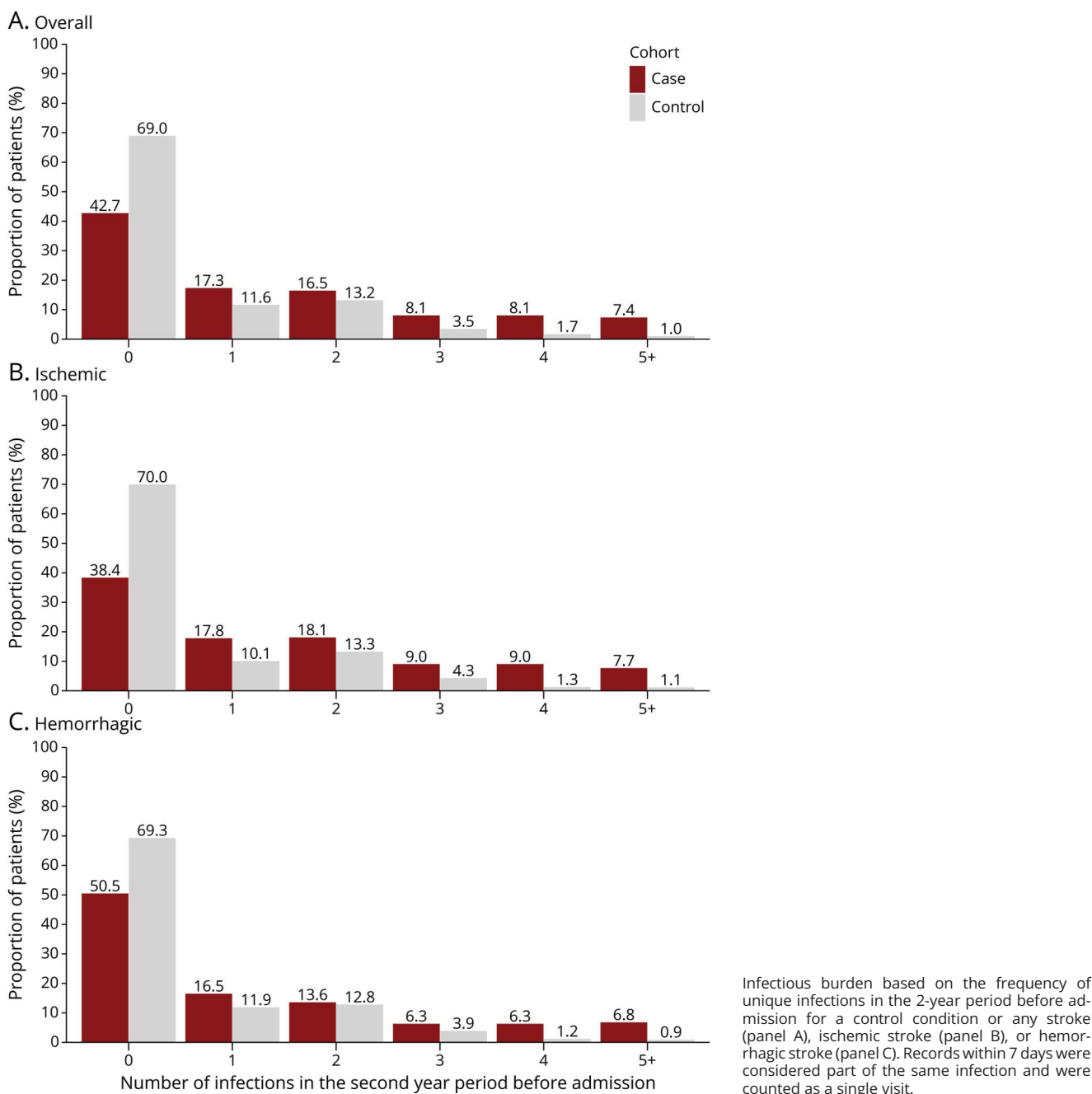
a significant association between recent vaccination and childhood stroke. Our results reinforce the role of infections as a risk factor for childhood stroke and the importance of infection prevention strategies (e.g., vaccination, hand washing) to potentially reduce the downstream risk of childhood stroke.

Rates of childhood stroke in our study (3.7 ischemic and 2.1 hemorrhagic events per 100,000) were greater than pooled estimates from systematic reviews (2.2 [min–max 0.9–7.9] ischemic and 1.5 [min–max 0.1–3.2] hemorrhagic events per 100,000).^{4,27,28} This may be explained by our use of comprehensive state-wide hospitalizations and death data to ascertain stroke. Similar to previous studies,^{28,29} the incidence of childhood stroke was highly age-related, following a reverse J-curve,³⁰ with decreasing incidence after infancy (28 days–1 year) before increasing slightly in late adolescence. Similar to previous research,³¹ male individuals had a significantly greater incidence of childhood stroke than female counterparts, with the greatest difference observed in those aged 28 days to 1 year. We also observed the incidence of childhood stroke was greater among individuals from areas of socioeconomic disadvantage. This finding aligns with previous studies^{8,32} and highlights the need to address social determinants of health from an early age to prevent adverse health outcomes, including childhood stroke.

During our study period, we observed an increase in the rate of childhood stroke (from ≈ 5 in 2017 to ≈ 7 events in 2023, per 100,000). Although it is difficult to ascertain the reason for this increase, we postulate that improved imaging, diagnosis, and coding practices for childhood stroke may have led to better documentation and recording of childhood stroke. Within Victoria, a second dedicated pediatric hospital (Monash Children's Hospital) opened in mid-2017, which may have also contributed to enhanced diagnosis and recording of childhood stroke. Other reasons might relate to the COVID-19 pandemic, with Victoria experiencing some of the longest periods of lockdown in the world.³³ In France, the

incidence of childhood stroke initially decreased during the COVID-19 pandemic (January 2020–June 2021), but slightly increased thereafter (July 2021–December 2022).³⁴ Although we did not observe a significant change in stroke incidence in our study during the COVID-19 pandemic (2020 and 2021), we did observe a similar increase in incidence in the post-pandemic period from 2022 onward. Whereas in the United States, the incidence of childhood stroke also declined during the COVID-19 pandemic and is thought to be because of widespread community infection prevention strategies, which reduce transmission of communicable diseases.³⁵ Further studies are needed to investigate temporal changes in childhood stroke incidence before, during, and after the COVID-19 pandemic.

A major finding of this study is that a recent infection in the previous 60 days was associated with 2-fold increased odds of childhood stroke. We also found that the risk of childhood stroke was the greatest in the 14-day period postinfection, associated with an 8.7-fold increased risk of ischemic stroke and 4.1-fold increased risk of hemorrhagic stroke. The association between recent infection and hemorrhagic stroke is novel and requires further investigation to ascertain potential causative mechanisms for this relationship. Whereas, our results for ischemic stroke align with previous literature. For example, in the VIPS study having a recent infection (in the past week) conferred a 6.3-fold increased risk of acute ischemic stroke compared with matched controls.⁸ Whereas, in another population-based study in the United States (1993–2007),³⁶ children with a health care visit for an infection in the past 3 days had an 8.3-fold increased risk of acute ischemic stroke than healthy matched controls. Although the authors did not identify an association between infections 4–90 days before stroke, the study was limited by the small sample size (102 cases). Our findings of an elevated risk of childhood ischemic stroke up to ≈ 180 days postinfection, and childhood hemorrhagic stroke up to ≈ 60 days postinfection, provide comprehensive evidence on these

Figure 1 Infectious Burden in the Previous 2-Year Period Among Cases of Childhood Stroke and Matched Controls, Overall and by Stroke Type

associations in a larger and more contemporary cohort. Further research is needed to validate these findings in other countries and data sets. Additional research is also needed to ascertain whether stroke is secondary to an abnormal host response, which is supported by emerging data from the Vascular Effects of Infection in Pediatric Stroke II study.³⁷

Infection in the 60 days before stroke was common, occurring in 2 in 5 patients. The most prevalent infections recorded among patients with childhood stroke were respiratory

infections, sepsis, meningitis/encephalitis, gastrointestinal tract infections, and urinary tract infections. Although these infections mostly align with previous case-control studies on this topic,^{36,38} we were unable to determine associations between certain neurotropic infections, previously implicated with childhood stroke, such as varicella zoster and herpes simplex infections.^{7,39} These infections had low prevalence in our cohort, possibly because of subclinical manifestations with reduced health care attendance, infection screening, and subsequent reporting in our government data sets.

Table 4 Association Between Recent Infection and Vaccination on the Risk of Childhood Stroke, Overall and by Stroke Type

	Cases of childhood stroke n (%)	Matched controls ^a n (%)	<i>p</i> Value	Matched analysis OR (95% CI)	Matched analysis with adjustment ^{a,b} OR (95% CI)
Combined	N = 571	N = 2,734			
Any infection in the previous 60 d					
No	332 (58.1)	2,225 (81.4)	<0.001	Ref	Ref
Yes	239 (41.9)	509 (18.6)		3.24 (2.65–3.96)	2.54 (1.97–3.27)
Any vaccination in the previous 42 d					
No	547 (95.8)	2,667 (97.6)	0.020	Ref	Ref
Yes	24 (4.2)	67 (2.5)		1.63 (1.00–2.65)	1.69 (0.92–3.09)
Ischemic stroke	N = 365	N = 1,734			
Any infection in the previous 60 d					
No	201 (55.1)	1,434 (82.7)	<0.001	Ref	Ref
Yes	164 (44.9)	300 (17.3)		4.15 (3.21–5.38)	3.39 (2.45–4.69)
Any vaccination in the previous 42 d					
No	346 (94.8)	1,683 (97.1)	0.029	Ref	Ref
Yes	19 (5.2)	51 (2.9)		1.65 (0.95–2.86)	1.90 (0.96–3.76)
Hemorrhagic stroke	N = 206	N = 1,000			
Any infection in the previous 60 d					
No	131 (63.6)	804 (80.4)	<0.001	Ref	Ref
Yes	75 (36.4)	196 (19.6)		2.36 (1.69–3.29)	1.90 (1.24–2.90)
Any vaccination in the previous 42 d					
No	201 (97.6)	975 (97.5)	0.95	Ref	Ref
Yes	5 (2.4)	25 (2.5)		0.98 (0.37–2.63)	0.93 (0.25–3.54)

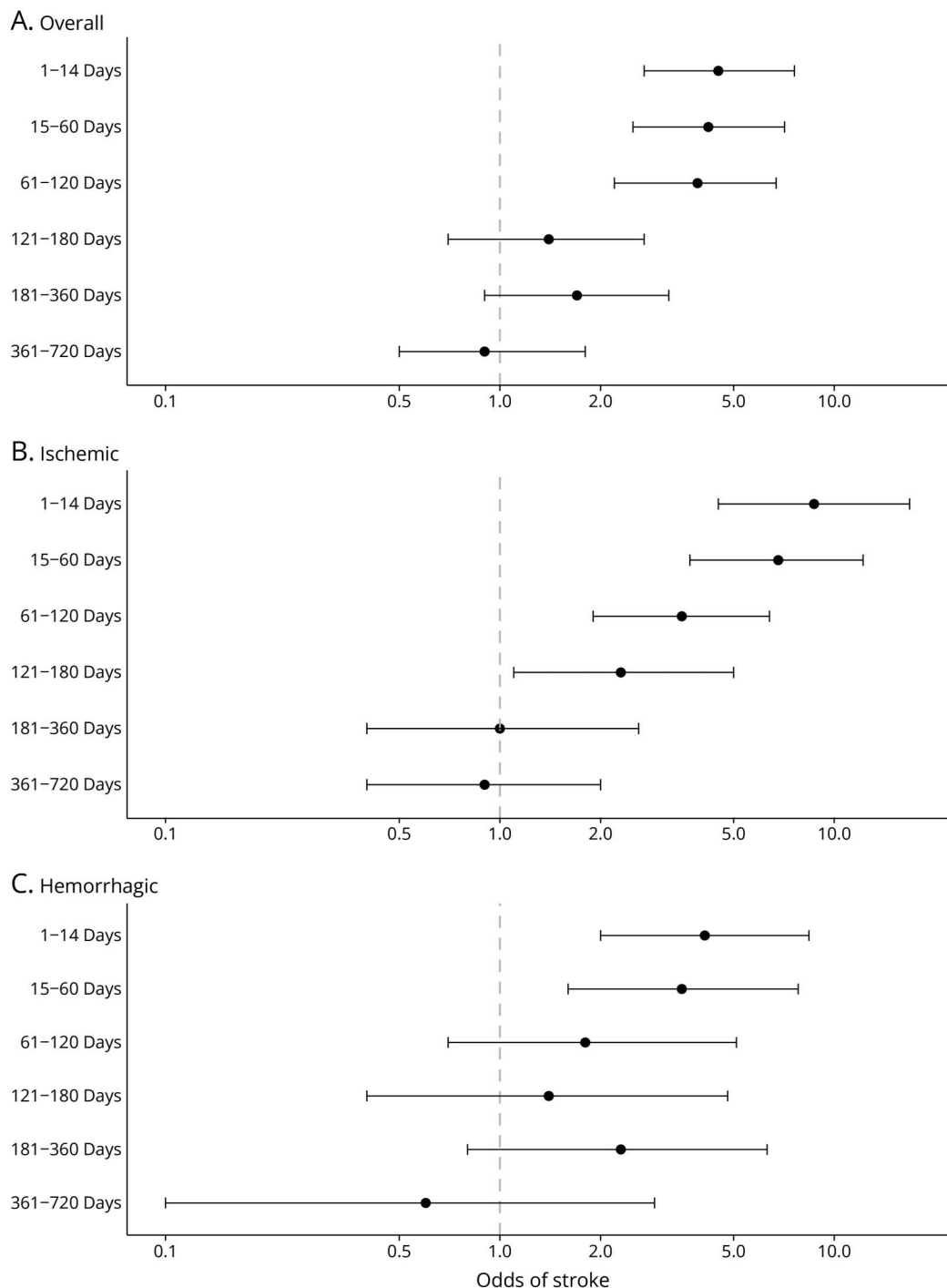
Abbreviation: OR = odds ratio.

^a Each case of childhood stroke matched with up to 5 controls, based on age (1-year bands), sex, quintiles of the Index of Relative Socioeconomic Advantage and Disadvantage, categories of childhood comorbidity score, and year of admission.^b Also adjusted for age (continuous variable), Indigenous status, born outside of Australia, and childhood comorbidity score.

Among the upper respiratory viruses identified in the 60-day period before stroke, COVID-19, RSV, pneumococcal disease, and influenza virus were the most common. These are among the most prevalent vaccine-preventable infections in the Australian population.⁴⁰ The significant association between recent RSV infection and childhood stroke is a novel and clinically relevant finding, given the growing awareness of the dangers of RSV infection on downstream vascular health.⁴¹ In a large cross-sectional study across 12 US states (2014/15–2022/23; N = 6,248), ≈1 in 4 patients aged ≥50 years who were hospitalized with RSV also experienced an acute cardiac event.⁴¹ Since early 2025, both state-funded nirsevimab, a long-acting monoclonal antibody preparation against RSV for infants and children younger than 2 years, and federally funded antenatal RSV vaccination for expectant mothers, have been available in Australia.⁴² Further research is needed to evaluate

the potential long-term effectiveness of routine childhood vaccination programs on downstream stroke incidence, using a larger sample size and longer follow-up period.

Childhood stroke did not seem to be associated with receiving a recent vaccination, acknowledging that many cell counts were too small to draw definitive conclusions. The US Centers for Disease Control study recently reported an increased risk of stroke in older adults after influenza vaccination,⁴³ although this association was subsequently shown to be inconsistent across flu seasons and age groups.⁴⁴ Further vaccine safety surveillance using a larger and more diverse sample may be needed to ensure there is no increased risk of ischemic stroke after influenza vaccination. Any possible risks associated with vaccination should be carefully weighed against the established benefits of influenza vaccination.⁴⁵

Figure 2 Sensitivity Analysis Examining Timing of Recent Infection and the Association With Childhood Stroke, Overall and by Stroke Type

Timing of recent infection and the association with any stroke (panel A), ischemic stroke (panel B), or hemorrhagic stroke (panel C). Reference category is no infection in the previous 720 days vs an infection in the following mutually exclusive look-back periods: 1-14, 15-60, 61-120, 121-150, 180-360, and 361-720 days. Each case of childhood stroke matched with up to 5 controls, based on age (1-year bands), sex, quintiles of the Index of Relative Socioeconomic Advantage and Disadvantage, categories of childhood comorbidity score, and year of admission. Also adjusted for age (continuous variable), Indigenous status, born outside of Australia, and childhood comorbidity score.

Strengths of this study include the population-based sample available for analysis, with up-to-date linked information from a range of important sources (e.g., hospitalizations, vaccinations, infections, and deaths). We also had data on the exact

timing of all vaccinations and hospital encounters to enable reliable ascertainment of past exposure. The notifiable infections register contained information on 76 confirmed infections with mandated reporting requirements in the state of

Victoria. We also supplemented the identification of infections with ICD-10-AM coded admissions for a range of common infections to maximize ascertainment.

There are limitations worth acknowledging for this study. First, the reliance on deidentified administrative claims records to identify childhood strokes may have increased the likelihood of misclassification of the outcome. A longer look-back period may have been needed to reliably differentiate incident and prevalent stroke events. Second, as the primary purpose of administrative data is for health care reimbursement, we did not have access to source medical records to confirm childhood stroke diagnoses and to extract more granular data on stroke causes, severity, and treatments provided. The lack of outpatient, primary care, and laboratory data also means that we have likely underestimated the incidence of childhood stroke and infections, particularly those subclinical in nature. As our data set was deidentified we were unable to corroborate infection histories and more precise onset dates through medical record review or interviews with patients or family members. However, evidence from VIPS indicates that only 55% of patients with a parental-reported infection had a confirmed positive infection in follow-up metagenomic next-generation sequencing.⁴⁶ This suggests that other pathogens may have been responsible for the infection among stroke patients, or that parental review overestimated the true prevalence of infections.⁴⁶

Finally, although our study included a large number of childhood strokes, we were likely underpowered to conclusively determine the effect of specific infections or vaccines on the risk of stroke. Particularly, the prevalence of varicella zoster and herpes simplex viruses was insufficient in our data set for comprehensive investigation. Further research, using a larger sample size, and more granular clinical data, are needed to determine the association of specific infections and vaccinations with childhood stroke. Nevertheless, our study provides contemporary and population-based evidence that is likely generalizable to other high-income countries with similar health care systems and practices.

Our findings provide new evidence on the epidemiology of childhood stroke and demonstrate the association of recent infections, such as respiratory infections, with childhood stroke. Clinicians should be aware of the growing evidence implicating common infections as risk factors for childhood stroke and actively promote preventive strategies, including timely vaccination, to mitigate this modifiable risk.

Acknowledgment

We acknowledge the Victorian Department of Health as the source of VAED, VEMD, VDI, and PHESS data, and the Commonwealth Department of Health as the source of Australian Immunisation Register data. We also acknowledge the Centre for Victorian Data Linkage for coordinating data custodian approvals, and regularly linking and transferring data each month. We thank members of the Vaccine Safety

Health Link Advisory Group and staff from the Centre for Health Analytics (Murdoch Children's Research Institute) for their data management assistance and advice.

Author Contributions

L.L. Dalli: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. M.T. Olaiya: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design. H.J. Morgan: major role in the acquisition of data. M.F. Kilkenney: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. M.C. Fahey: drafting/revision of the manuscript for content, including medical writing for content. M.T. Mackay: drafting/revision of the manuscript for content, including medical writing for content. D.A. Cadilhac: drafting/revision of the manuscript for content, including medical writing for content. T.-Y. Kuo: analysis or interpretation of data. S.L. Gall: drafting/revision of the manuscript for content, including medical writing for content. H. Clothier: major role in the acquisition of data. J.A. Boyle: drafting/revision of the manuscript for content, including medical writing for content. J. Mallard: major role in the acquisition of data. D. Hennessy: major role in the acquisition of data. J. Buttery: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design.

Study Funding

The establishment of the Vaccine Safety Health Link has been funded by the Department of Health, Victoria, The Royal Children's Hospital Foundation and the Murdoch Children's Research Institute. This study was funded by grants from the Jack Brockhoff Foundation (Early Career Researcher Medical Grant 5119-2023), National Health and Medical Research Council Synergy Grant (GNT1182071), and Monash University.

Disclosure

L.L. Dalli reports membership of the Victorian Cardiovascular Research Network Steering Committee and Editorial Board of *Stroke* (American Heart Association), and has received fellowship support from the National Heart Foundation of Australia (107125). M.T. Olaiya reports Editorial Board membership of *Stroke* (American Heart Association). M.F. Kilkenney reports membership of advisory committees for the Stroke Foundation, the Australian Institute of Health and Welfare, and the Australian Stroke Clinical Registry (AuSCR); and has received fellowship support from the National Heart Foundation of Australia (105737). D.A. Cadilhac reports executive leadership of the AuSCR and Australian and New Zealand Stroke Organization; and receiving restricted grants from Boehringer Ingelheim and Amazon Web Services outside the submitted work. S.L. Gall reports funding from the National Heart Foundation of Australia (108524) and National Health

and Medical Research Council (GNT1182071); and funding from the Medical Research Future Fund (GNT2015976) outside the submitted work. All other authors report no disclosures relevant to the manuscript. Go to [Neurology.org/N](https://www.neurology.org/N) for full disclosures.

Publication History

Received by *Neurology*® July 24, 2025. Accepted in final form December 17, 2025. Submitted and externally peer reviewed. The handling editor was Associate Editor Courtney Wusthoff, MD, MS.

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